

Recursive Binary Words as Dimension-Bounded Temporal Memory Witnesses

Youfu Wang

Independent Researcher, Anyang, China

How much internal memory is required for an autonomous device to emit a specified temporal word when the same binary instrument is applied at every step? We answer this question for the dyadic finite Thue–Morse words $T_n = \mu^n(0)$, with $\mu(0) = 01$ and $\mu(1) = 10$. If the stopping length is external and no clock, index, or round label enters the device, their exact deterministic classical complexity is $DC(T_n) = 3 \cdot 2^{n-2} = 3|T_n|/4$ for every $n \geq 2$. At lengths eight and sixteen we give machine-auditable global certificates for stationary two-state classical generators and explicit stationary qubit instruments. For $T_3 = 01101001$ the certified classical and quantum thresholds are $P_C < 0.03 < P_Q$ with $P_Q > 0.0414$; for $T_4 = 0110100110010110$ they are $P_C < 0.0005 < P_Q$ with $P_Q > 0.00069$. These establish equal-dimension temporal separations at two recursive scales. Under a registered reset-noise model for the length-eight test, the separation persists up to a per-round noise rate of $8.5449\ldots\%$; exact binomial calculations give concrete finite-sample requirements. The result is a dimension-bounded temporal-memory witness, not a device-independent statement and not evidence for hidden quantum states.

1 Introduction

Temporal correlations turn internal dimension into an operational memory resource. In the simplest stationary scenario, one finite-dimensional system is subjected repeatedly to the same binary-output instrument [1]. A device with insufficient memory cannot generate every requested word deterministically, and classical and

Youfu Wang: youfuwang888@gmail.com

quantum models of the same dimension can assign different probabilities to the same temporal event [2, 3]. This provides a particularly lean setting for dimension-bounded tests: there are no measurement choices and no time-dependent controls inside the abstract model.

The recursive Thue–Morse words are a natural stress test for autonomous memory. Although the infinite sequence is 2-automatic [4], the standard automaton reads the binary digits of the time index. Here the device receives no index or clock. It must encode progress through a finite target word in its internal state while the same instrument is repeated (Fig. 1). Thus two-automaticity does not imply two-state autonomous generation.

Contributions and assumptions. Our contributions are fourfold.

1. We prove an exact family theorem, $DC(T_n) = 3|T_n|/4$, for all dyadic finite Thue–Morse words.
2. For T_3 and T_4 , we certify global upper bounds on every stationary two-state classical generator.
3. We provide explicit stationary qubit instruments above both bounds. The quantum probabilities are verified by outward interval arithmetic.
4. We give a frozen noise model and exact-binomial power calculations for an implementation-oriented test.

The separation assumes a stationary two-dimensional memory, a binary instrument, and the absence of a hidden clock. These are trusted-model assumptions. The result is neither device independent nor a claim about a microscopic ontology.

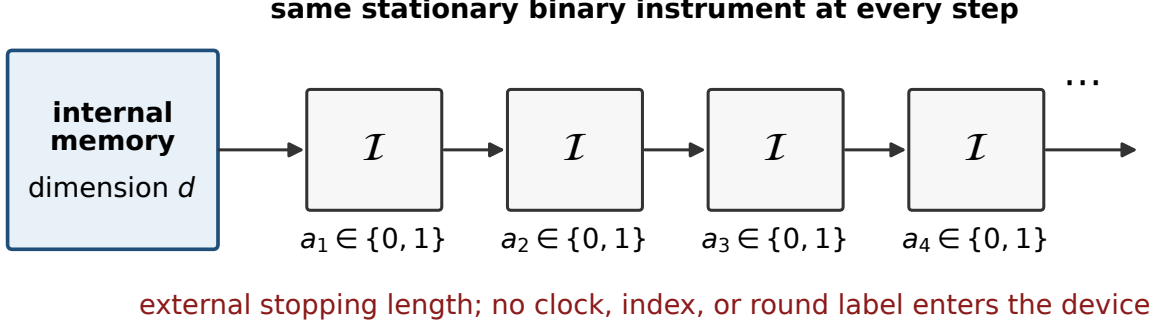


Figure 1: Operational task. A memory of fixed dimension d is subjected to the same stationary binary instrument $\mathcal{I}$ at every step. The experimenter stops after a prescribed external length, but no clock, index, position, or round label enters the device.

The closest operational study is Ref. [1], which introduced the deterministic complexity used here and numerically surveyed classical words through length ten and quantum words through length seven. Our contribution is word-specific and certificate-based: an exact recursive family theorem, two certified recursive scales, and reproducible global classical bounds. Recent comparisons of one-bit and one-qubit finite-state generators use different temporal scores and delayed measurement settings [5]. Other temporal dimension witnesses [2, 3, 10, 11, 6], hidden-memory tests [12], and quantum stochastic simulators [7, 8] address related memory questions under different operational tasks.

2 Stationary temporal generators

Let $w = a_1 \cdots a_L \in \{0, 1\}^L$. A classical generator of dimension d is specified by nonnegative $d \times d$ matrices T_0, T_1 satisfying

$$(T_0 + T_1)\mathbf{1} = \mathbf{1} \quad (1)$$

and an initial row distribution π . Its word probability is

$$P_C(w) = \pi T_{a_1} \cdots T_{a_L} \mathbf{1}. \quad (2)$$

The matrices jointly encode the emitted symbol and the memory transition.

A quantum generator of dimension d is a density operator ρ and a binary instrument $\{\mathcal{I}_0, \mathcal{I}_1\}$, where each $\mathcal{I}_a$ is completely positive and trace nonincreasing and $\mathcal{I}_0 + \mathcal{I}_1$ is trace preserving. Its

word probability is

$$P_Q(w) = \text{Tr}[\mathcal{I}_{a_L} \circ \cdots \circ \mathcal{I}_{a_1}(\rho)]. \quad (3)$$

Both models are stationary: the same pair of maps is used at every step. The stopping length L is imposed externally.

A deterministic classical generator has one outgoing symbol-transition from every reachable state. Along a finite run its state path is a tail followed by a cycle. The deterministic complexity $\text{DC}(w)$ is the minimum number of states in such a generator that emits w before the external stop [1].

3 Exact deterministic complexity

Define $\mu(0) = 01$, $\mu(1) = 10$, and

$$T_n = \mu^n(0), \quad N = |T_n| = 2^n. \quad (4)$$

Theorem 1 (Finite Thue–Morse complexity). *For every $n \geq 2$,*

$$\boxed{\text{DC}(T_n) = 3 \cdot 2^{n-2} = \frac{3N}{4}}. \quad (5)$$

The same formula holds for the bitwise complement $\overline{T_n}$.

The proof uses the longest repeated factor at a fixed displacement. For a finite word $w = w_0 \dots w_{N-1}$ define

$$M(w) = \max\{\ell : q \geq 1, s \geq 0, s + q + \ell \leq N, \\ w_{s+j} = w_{s+q+j} \text{ for } 0 \leq j < \ell\}. \quad (6)$$

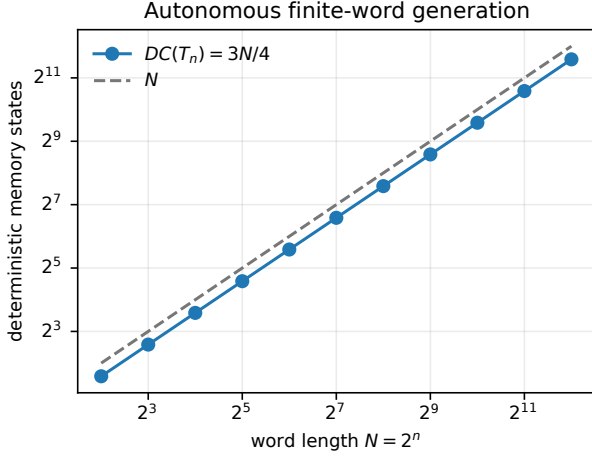


Figure 2: Exact deterministic memory of T_n . The standard two-state automatic description is not available here because it requires the binary digits of the time index as input.

For an odd displacement, no four consecutive symbols of the infinite Thue–Morse word can agree: even-start and odd-start length-four factors belong to disjoint finite sets. For an even displacement $2r$, the identities $t_{2i} = t_i$ and $t_{2i+1} = 1 - t_i$ duplicate every equality indicator at displacement r . Iterating this desubstitution and checking the terminal words T_1, T_2, T_3 gives $M(T_n) \leq N/4$. Conversely,

$$T_n = A \bar{A} \bar{A} A, \quad A = T_{n-2}, \quad (7)$$

so the prefix and suffix agree for $N/4$ symbols and $M(T_n) = N/4$.

If a tail-cycle generator has d states, then the final $N - d$ outputs repeat an earlier block at the cycle displacement. Hence $N - d \leq M(T_n) = N/4$, so $d \geq 3N/4$. A cycle containing the first $3N/4$ symbols reproduces the final block A in Eq. (7), proving attainability. Appendix A gives the finite-boundary details. Figure 2 shows the resulting linear memory law.

4 Certified dimension-two separation

We first fix the length-eight target

$$w = T_3 = 01101001. \quad (8)$$

Theorem 1 gives $DC(w) = 6$, so both dimension-two models operate strictly below the deterministic threshold.

4.1 Exact-rational classical upper bound

For $d = 2$, the initial distribution can be chosen pure because Eq. (2) is linear in π . Relabelling the two internal states then fixes $\pi = (1, 0)$ without loss of generality. The eight nonnegative entries of T_0, T_1 , subject to the two normalization equations in Eq. (1), leave six independent variables on a product of two simplices.

We cover this domain with rational boxes. For every box, each independent matrix entry is replaced by its rational upper endpoint, while the dependent entry in each stochastic row is bounded by one minus the sum of the rational lower endpoints. Positivity of all matrices makes the resulting forward propagation an upper bound on Eq. (2). Bisecting the widest coordinate produces a nested cover. After 130000 bisections, the largest exact bound over all 130001 queued boxes is

$$\frac{16282708641}{549755813888} = 0.029618074479 \dots < \frac{3}{100}. \quad (9)$$

Because the queued boxes still cover the full feasible domain, Eq. (9) is a global certificate rather than a variational estimate. Appendix B states the algorithm and its coverage invariant.

4.2 Explicit qubit instrument

Let $\rho = |\psi\rangle\langle\psi|$ with

$$|\psi\rangle = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}, \quad \theta = -0.2880903478104165, \quad (10)$$

and define

$$K_0 = \begin{pmatrix} -0.261100885994 & -0.263277264622 \\ 0.894987283427 & -0.279312440034 \end{pmatrix}, \quad (11)$$

$$K_1 = R(\phi) \sqrt{I - K_0^\top K_0}, \quad (12)$$

$$\phi = 2.1351675998196673,$$

where $R(\phi)$ is the real two-dimensional rotation. The residual $I - K_0^\top K_0$ is positive definite, and $K_0^\dagger K_0 + K_1^\dagger K_1 = I$ holds by construction. Outward interval evaluation at 80 decimal digits gives

$$\begin{aligned} &P_Q(01101001) \\ &= \|K_1 K_0 K_0 K_1 K_0 K_1 K_0 |\psi\rangle\|^2 \\ &\in [0.04146981659604025 \dots, \\ &\quad 0.04146981659604026 \dots], \end{aligned} \quad (13)$$

Proposition 2 (Certified temporal separation).
Under the stationary, clock-free model of Sec. 2,

$$\begin{aligned} \sup_{\substack{\text{classical} \\ d=2}} P(01101001) &< \frac{3}{100} \\ &< 0.0414 \\ &< P_Q^{\text{explicit}}(01101001). \end{aligned} \quad (14)$$

4.3 Second certified scale

The same comparison can be made for

$$w' = T_4 = 0110100110010110.$$

An outward-rounded branch-and-bound cover is bisected until every remaining box lies below $1/2000$. Each endpoint in the final 176112-box cover is then converted to its exact binary rational value, and every box bound is recomputed using exact rational arithmetic. The exact maximum is $0.000499999289033821 \dots < 1/2000$. An independently parameterized, exactly normalized qubit instrument, stated in Appendix D, has an outward-interval probability $0.000696451992086557 \dots > 0.00069$. Therefore

$$\sup_{\substack{\text{classical} \\ d=2}} P(T_4) < \frac{1}{2000} < 0.00069 < P_Q^{\text{explicit}}(T_4). \quad (15)$$

All displayed quantum values are feasible lower bounds, not claims of quantum optimality. Numerical classical constructions reach $0.016434148019 \dots$ for T_3 and $9.488079226956 \times 10^{-5}$ for T_4 . Figure 3 shows these constructions only to display the remaining gaps inside the certified classical intervals.

5 Conditional robustness and finite samples

To make the statistical scale explicit, we freeze one trusted noise model. At each round, with probability $1 - \eta$ the ideal outcome map is applied; with probability η , the device outputs a fair random bit and resets the qubit to $I/2$. This is a registered implementation model, not an arbitrary-noise or device-independent statement. Exact channel propagation gives a crossing of the classical threshold at

$$\eta_c = 0.085449325714 \dots \quad (16)$$

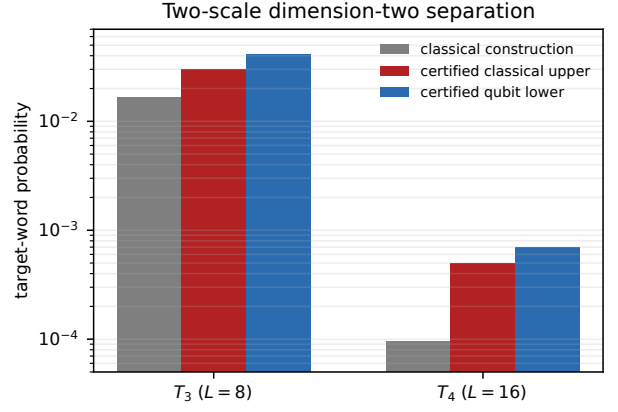


Figure 3: Certified equal-dimension separation at two recursive scales. Red bars are rigorous global classical upper thresholds; blue bars are rigorous explicit-qubit lower bounds. Gray bars are feasible classical constructions and do not define the bounds.

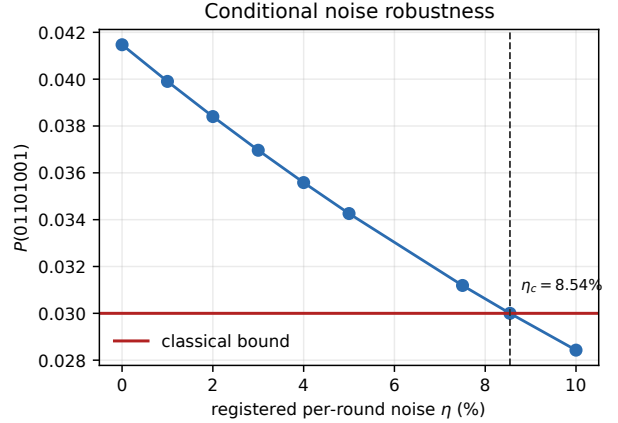


Figure 4: Target-word probability under the frozen per-round reset-noise model. Robustness outside this model is not claimed.

At $\eta = 0.02$, the target probability is $0.038402921993 \dots$ (Fig. 4).

For m independent length-eight trials, let X be the target-word count. Testing the conservative null $p_0 = 0.03$ with an exact binomial upper-tail test, a one-sided five-sigma type-I error and at least 90% power require $m = 9957$ ideal trials, rejecting for $X \geq 388$. At $\eta = 0.02$, the corresponding design is $m = 17979$ and $X \geq 658$. The achieved type-I errors are 2.85×10^{-7} and 2.82×10^{-7} , respectively. These figures do not include uncertainty in a physical dimension calibration or violations of stationarity.

6 Discussion

What is certified. The result separates stationary two-state classical generators from explicit stationary qubit instruments on two preselected members of one recursive family. Both classical bounds are global within the stated model and end in exact rational verification. The entire dyadic family has an exact deterministic memory law.

What is not certified. A physical implementation would have to justify the dimension-two null and exclude a hidden clock, time-dependent control, side channels, and postselection. The noise calculation is conditional. The result does not identify hidden quantum states, does not modify the Born rule, and does not claim that the Thue–Morse substitution is a microscopic law of nature.

Relation to word complexity. Automatic complexity studies the smallest automata recognizing individual finite strings [9]. Our deterministic quantity belongs to a different operational model: autonomous emission by one repeated instrument with an external stop. The distinction is essential for recursive automatic sequences, whose compact index-reading descriptions otherwise look like a contradiction to Theorem 1.

Open problems. The most important next step is an asymptotic probabilistic separation for the family T_n . The two certified scales reported here establish recurrence, not an asymptotic law. On the classical side, sharper exact bounds could be obtained with polynomial optimization or support elimination. On the quantum side, recursive instruments or matrix-product constructions may expose a scalable ratio. Experimentally, an independently dimension-bounded platform and a preregistered stationarity audit would turn Proposition 2 into a direct temporal memory test.

7 Conclusion

Recursive complement structure can be simple for an index-reading automaton yet memory intensive for an autonomous clock-free generator. Finite Thue–Morse words make this distinction

exact: $\text{DC}(T_n) = 3|T_n|/4$. At the first two dyadic lengths beyond the prior quantum survey, global classical certificates and explicit qubit instruments give strict equal-dimension separations. The construction supplies a compact, reproducible temporal-memory witness with a clearly delimited experimental route.

Data and code availability. All source code, exact certificate outputs, numerical witness data, plotted data, and a one-command reproduction script will be archived in a public repository and assigned a persistent DOI before submission.

Author contributions and AI-use disclosure. Y.W. originated the recursive-closure research direction, defined the research objective, and is the sole human author responsible for the work. OpenAI Codex was used under the author’s direction for bibliographic research, code and calculation implementation, theorem and manuscript drafting, figure production, and formatting. The author will review and approve all final claims, code, and submitted materials and assumes full responsibility for them.

Competing interests. The author declares no competing interests.

A Full deterministic proof

Let $t = t_0 t_1 \dots$ be the infinite Thue–Morse word. It obeys

$$t_{2i} = t_i, \quad t_{2i+1} = 1 - t_i. \quad (17)$$

Lemma 3 (Odd displacements). *Two factors of t separated by an odd displacement agree on at most three consecutive symbols.*

Proof. Every length-four factor beginning at an even position is the image under μ of a length-two factor and belongs to

$$\{0101, 0110, 1001, 1010\}.$$

Every length-four factor beginning at an odd position is fixed by a length-three preimage factor and the one-symbol offset, and belongs to

$$\{0010, 0011, 0100, 1011, 1100, 1101\}.$$

The sets are disjoint. An odd displacement exchanges parity, so a four-symbol match is impossible. $\square$

Lemma 4 (Even-displacement desubstitution). *If $q = 2^s u$ with u odd, a matching run at displacement q has length at most 2^s times the longest matching run at odd displacement after s desubstitutions.*

Proof. For displacement $2r$, Eq. (17) gives

$$[t_{2i} = t_{2i+2r}] = [t_{2i+1} = t_{2i+1+2r}] = [t_i = t_{i+r}].$$

Thus the equality-indicator word for $2r$ is obtained from that for r by duplicating every symbol. A run of ones can grow by at most a factor two. Iterating s times proves the claim. Restricting to the finite prefix T_n can only truncate a run. $\square$

For $n - s \geq 4$, the preceding lemmas bound a run by $3 \cdot 2^s < 2^{n-2}$. Direct inspection of T_3, T_2, T_1 gives odd-shift maxima 2, 1, 0, respectively, and the same upper bound in all terminal cases. Hence $M(T_n) \leq 2^{n-2}$. Equation (7) attains this value.

Any deterministic trajectory on d states repeats a state among the first $d + 1$ visited states. Its emitted word is therefore a tail followed by a cycle with total tail-plus-cycle length at most d . If it emits N symbols, the last $N - d$ symbols match the block one cycle earlier, so $N - d \leq M(T_n)$. This yields $d \geq 3N/4$. The cycle construction following Eq. (7) attains equality. Complementation preserves every equality relation, completing the proof of Theorem 1.

B Exact-rational classical certificate

For each source state, list the four joint outcome-destination probabilities in the order

$$(0, 0), (0, 1), (1, 0), (1, 1).$$

The first three are independent variables $x_1, x_2, x_3 \geq 0$ with $x_1 + x_2 + x_3 \leq 1$; the fourth is $1 - x_1 - x_2 - x_3$. Repeating this for the second source state gives six variables.

For a rational box $B = [\ell, u]$, feasibility tightening replaces each upper endpoint by

$$u_i \leftarrow \min \left(u_i, 1 - \sum_{j \neq i} \ell_j \right)$$

within its simplex. The independent transition entries are bounded by u_i , and the dependent entry by $1 - \sum_i \ell_i$. Call the resulting nonnegative

entrywise upper matrices $\hat{T}_0(B), \hat{T}_1(B)$. Starting from the pure initial state, positive matrix multiplication yields an upper vector after every symbol. The sum of its entries is therefore an upper bound $U(B)$ on the target probability throughout B .

The algorithm begins with the two full simplices, removes infeasible boxes, and bisects a widest rational interval at its midpoint. Replacing one box by its feasible children preserves coverage of the parameter domain. Consequently, at every iteration,

$$\sup_{\theta} P_C(w; \theta) \leq \max_{B \text{ queued}} U(B).$$

All endpoints, propagation steps, comparisons used for the final bound, and the final maximum are exact **Fraction** operations. Floating-point values are used only to prioritize the queue and cannot affect coverage or the final inequality. After 130000 processed boxes the exact maximum is Eq. (9). Its distance below $3/100$ is exactly

$$\frac{3}{100} - \frac{16282708641}{549755813888} = \frac{5249144391}{13743895347200} > 0.$$

For T_4 , a faster first stage uses IEEE-754 operations with every multiplication, addition, and simplex tightening rounded outward via **nextafter**. Best-first bisection leaves 176112 boxes whose largest outward upper bound is 0.000499999289033824.... In a separate second stage, every endpoint of every remaining box is converted to its exact binary rational value and the full length-16 propagation is recomputed with **Fraction** operations. No box reaches $1/2000$; the largest exact bound is 0.000499999289033821.... Because every processed box is replaced by its feasible children, the final queue retains the same coverage invariant as above. The exact numerator and denominator are included in the machine-readable certificate rather than printed here.

C Classical feasible construction

The best classical value found numerically lies on the support

$$T_0 = \begin{pmatrix} a & 0 \\ 1-a & a \end{pmatrix}, \quad T_1 = \begin{pmatrix} b & c \\ 0 & 0 \end{pmatrix},$$

$$a + b + c = 1,$$

with initial state $(0, 1)$. The target probability is

$$P(a, b) = ab(a-1)^2(a^2 + 2ab - 2a - b + 1) \times (2a^2 + 3ab - 4a - 2b + 2). \quad (18)$$

The feasible stationary point has

$$a = 0.142664084620\dots, \\ b = 0.326162782878\dots,$$

and gives $P = 0.016434148019\dots$. This construction is a lower bound on the classical optimum only and is not used in Proposition 2.

D Quantum witness verification

With one Kraus operator per outcome,

$$\mathcal{I}_a(\rho) = K_a \rho K_a^\dagger.$$

The archived machine-readable parameters retain the displayed decimal definitions. Positivity of $I - K_0^\top K_0$ is checked by interval arithmetic. The square-root completion makes trace preservation exact by definition, while an independent 80-digit outward-interval computation certifies $P_Q > 0.0414$.

For T_4 , the same exactly normalized parameterization uses

$$\theta = -0.35791546499923926, \\ \phi = 2.1733161717864493.$$

$$K_0 = \begin{pmatrix} -0.293850518688 & -0.219854695403 \\ 0.862588464315 & -0.314384115846 \end{pmatrix}.$$

$$K_1 = R(\phi) \sqrt{I - K_0^\top K_0}.$$

The residual eigenvalues are $0.1119890568958\dots$ and $0.9104304976137\dots$. An independent 80-digit outward-interval computation certifies $P_Q(T_4) > 0.00069$.

E Exact-binomial design

For $X \sim \text{Binomial}(m, p)$, the rejection threshold is the smallest integer k satisfying

$$\Pr_{p=0.03}[X \geq k] \leq \alpha.$$

Power is $\Pr_{p_1}[X \geq k]$. We search integer m and k directly, without a normal approximation. Table 1 records the frozen designs.

Table 1: Exact-binomial designs with at least 90% power.

Condition	α	m	k	power
ideal	10^{-3}	4877	185	0.90009
$\eta = 0.02$	10^{-3}	8788	315	0.90000
ideal	$2.87 \cdot 10^{-7}$	9957	388	0.90018
$\eta = 0.02$	$2.87 \cdot 10^{-7}$	17979	658	0.90018

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